- **QuestDB** is a relational column-oriented database that can perform real-time analytics on time series data. It works with SQL and some extensions to create a relational model for time series data. It supports relational and time-series joins, which helps in correlating the data.
- **OpenTSDB** is a scalable time series database that has been written on top of HBase. It is capable of storing trillions of data points at millions of writes per second. It has a time-series daemon (TSD) and command-line utilities. TSD is responsible for storing data in or retrieving it from HBase. You can talk to TSD using HTTP API, telnet, or a simple built-in GUI. You need tools like flume, collectd, vacuumetrix, etc, to collect data from various sources into OpenTSDB.

Cloud native TSDBs

Cloud hyperscalers like Azure, AWS and Google offer time series databases and analytics services as part of their cloud portfolio. AWS Timestream is a serverless time series database service that is fast and scalable. It is used majorly for IoT applications to store trillions of events in a day and is 1000 times faster with 1/10th the cost of relational databases. Using its purpose-built query engine, you can query recent and historical data simultaneously. It provides multiple built-in functions to analyse time series data to find useful insights.

Microsoft Azure Time Series Insights provides a time series analytics engine. For data ingestion there are Azure IoT Hub and Event Hub services. To analyse cloud infrastructure and time series streams, these cloud vendors offers a range of native tools such as AWS CloudWatch, Azure Monitor, Amazon Kinesis, etc. END

By: Narasimha Sekhar Kakaraparthi

The author has more than 27 years of software industry experience in product development in the hybrid cloud, telecommunications, and manufacturing domains. He is currently working as principal architect for virtual desktops and end user services at Wipro Technologies. He is a senior member of Wipro's eminent architects group called Distinguished Member of Technical Staff (DMTS).

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His idea got written about in newspapers in Bangalore (as the city was known then), Karachi, and elsewhere. He recalls that a news story in the *Deccan Herald* drew attention to his contributions, as did another in *The Dawn.*

He then started getting responses from a series of obviously foreign names. He learnt why this was happening only when he asked what had triggered their interest — it was the newspaper articles.

Hacking for learning

Doc Partha's ebook on LaTeX — the free and open source software system for document preparation — has been used in more than 65 countries, spanning six continents. His goal is to promote "hacking for learning" through his books and publications, which he shares under a Creative Commons (CC-BY-SA) license. The latest version of this LaTeX generated document is available via *https://drpartha.org.in/.*

He takes pride in being a "regular supporter, promoter and practitioner of FOSS," and has contributed to international efforts to build this volunteer-driven system. His contributions feature in major GNU/Linux 'distros' (or distributions) worldwide, and have been translated into nine languages — Japanese, Dutch, French, Turkish, Russian, Spanish, German, Czech and Bulgarian.

Among other feathers in his cap are being a senior member of IEEE (USA) and a life fellow of IETE (India). He was twice decorated by the IEEE (USA) with international awards, for outstanding contributions to the profession.

But that's not all. In his spare time, he provides free advice and technical assistance for making handicapped-accessible buildings, structures and facilities. He is a serious promoter of eco-friendly, vermiculture-based organic farming, and an active campaigner and volunteer for organ donation.

Doc Partha has authored several research publications, popular articles and, in their time, educational CD-ROMs. He was an editor of the international journal 'Engineering Applications of Artificial Intelligence' (Elsivier, UK), and a reviewer for the Practex Journal, published by the TeX Users Group.

He recalls his "begging" for a book on Linux, by writing to people whom he "didn't even know." The book that Thaths sent him is still on his table. "I'm still learning from this book," says the 72-year old.

"It has been a long and interesting journey. Even today, I keep learning something new every day," he notes. END

By: Frederick Noronha

The author is an independent journalist and publisher, and has written on FOSS for almost two decades.